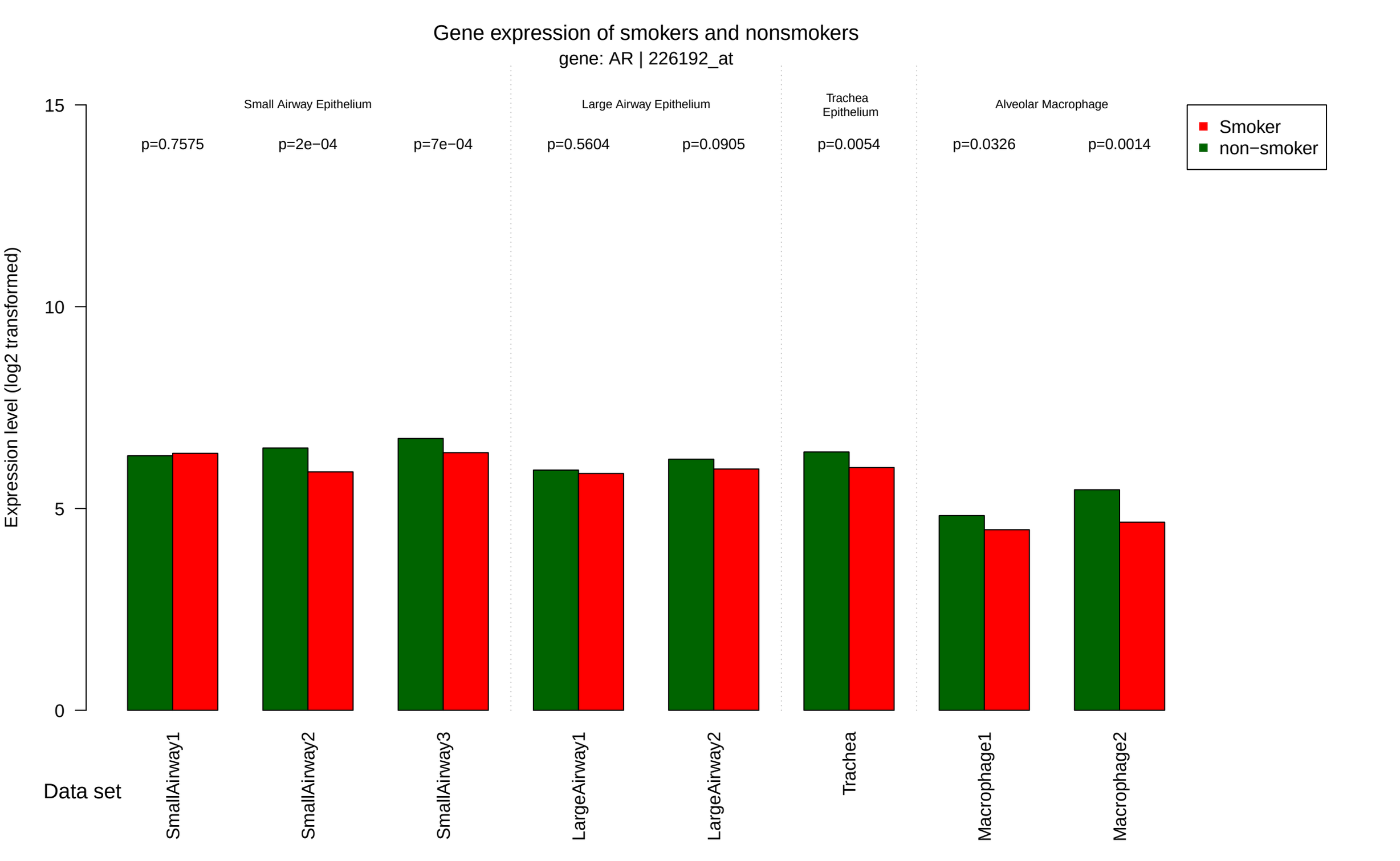
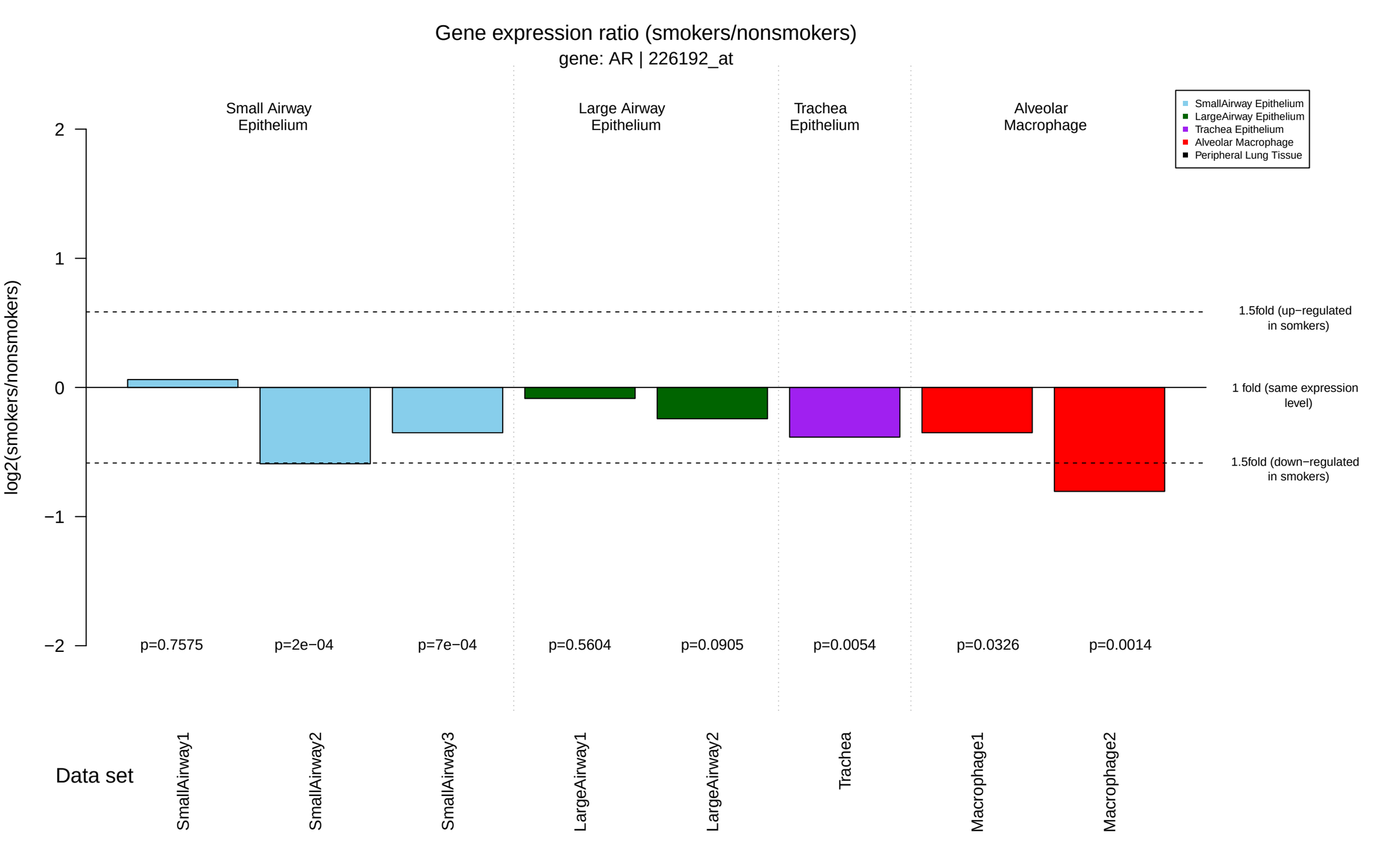
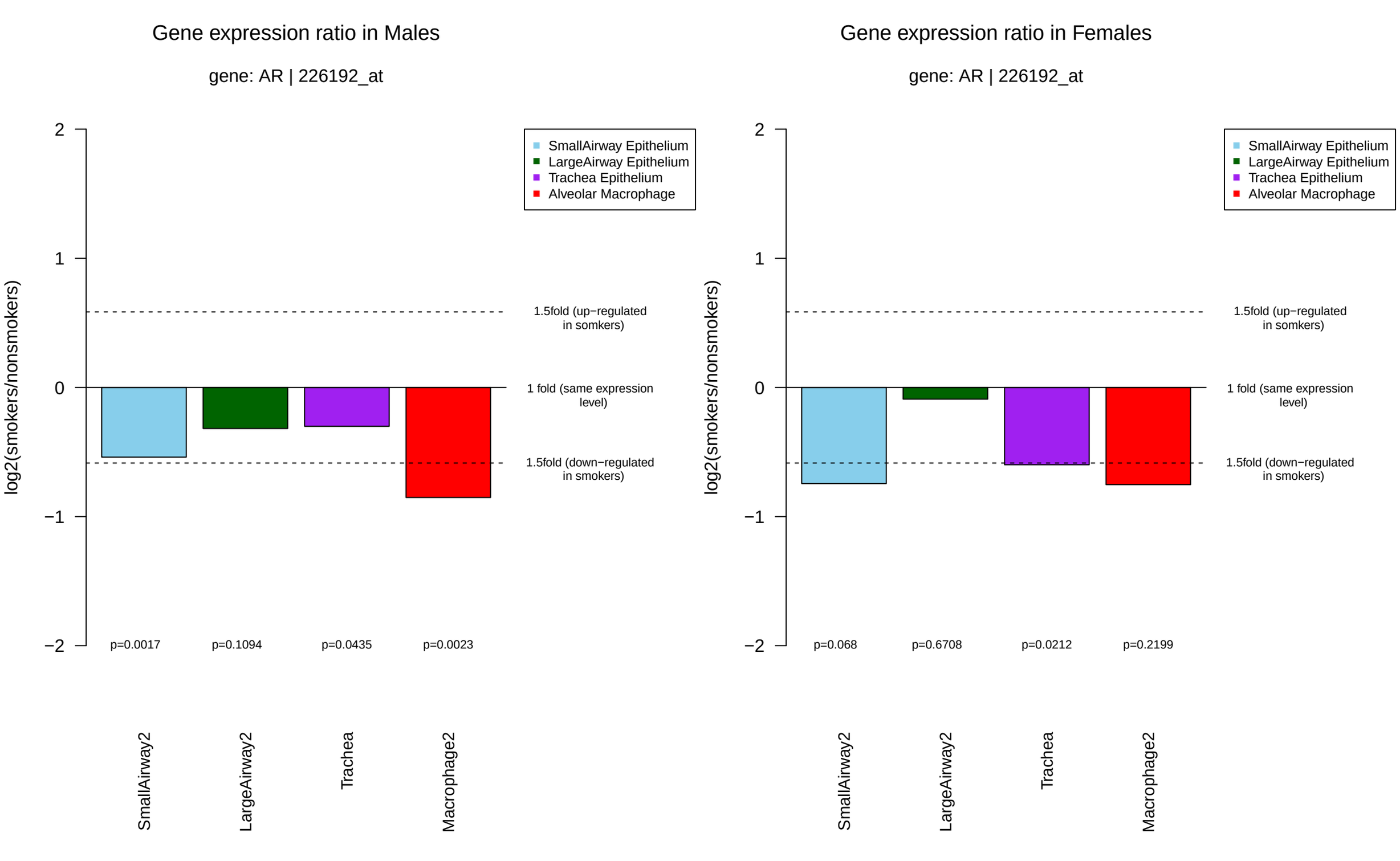
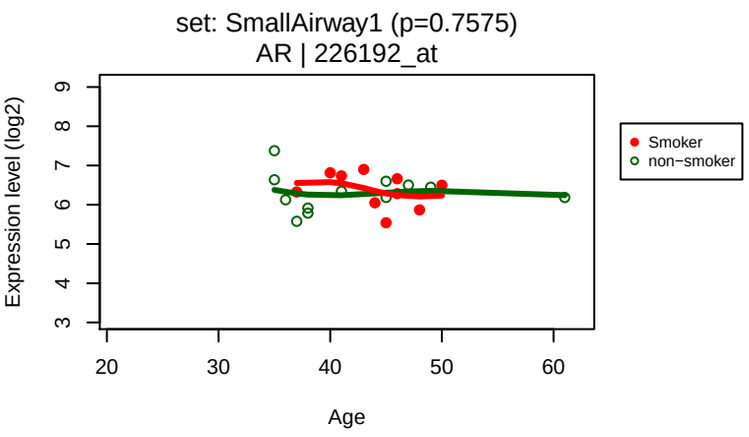
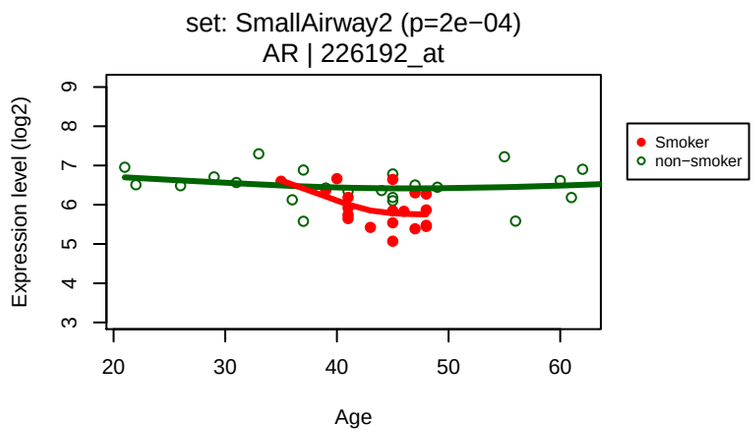
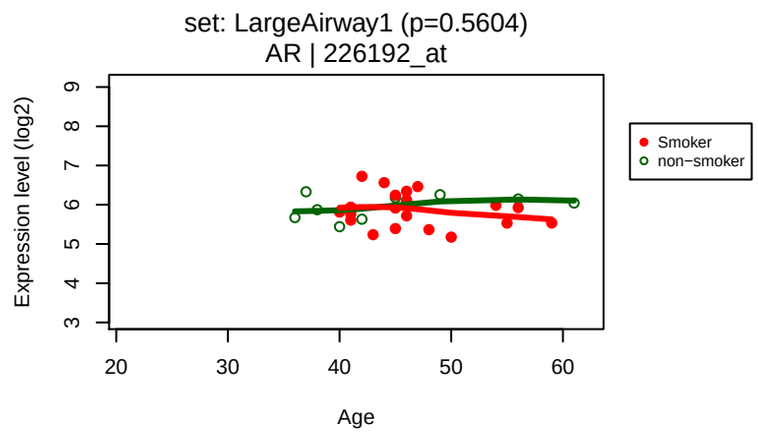

Gene expression of smokers and nonsmokers
gene: AR | 226192_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7575
p=2e−04
p=7e−04
p=0.5604
p=0.0905
p=0.0054
p=0.0326
p=0.0014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

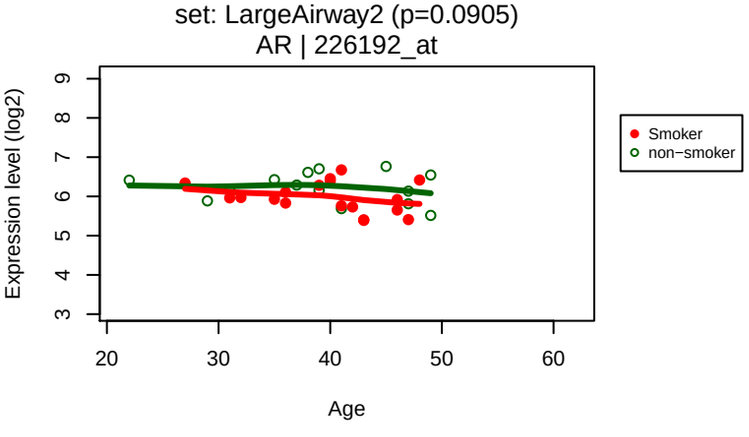

Gene expression ratio (smokers/nonsmokers)
gene: AR | 226192_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7575
p=2e−04
p=7e−04
p=0.5604
p=0.0905
p=0.0054
p=0.0326
p=0.0014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

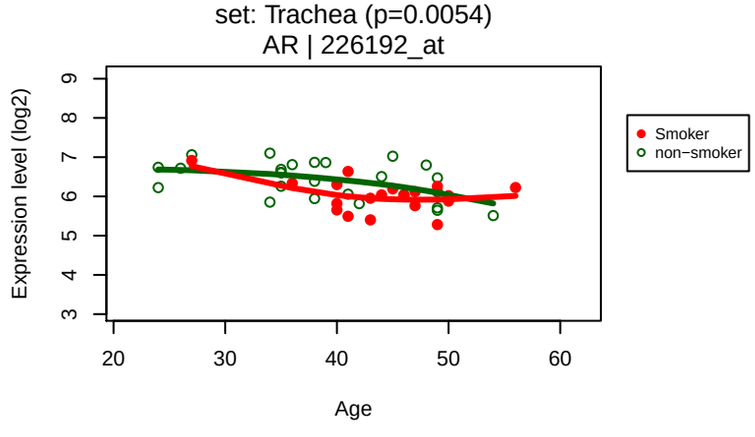

Gene expression ratio in Males
gene: AR | 226192_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.1094
p=0.0435
p=0.0023
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AR | 226192_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.068
p=0.6708
p=0.0212
p=0.2199
SmallAirway2
LargeAirway2
Trachea
Macrophage2

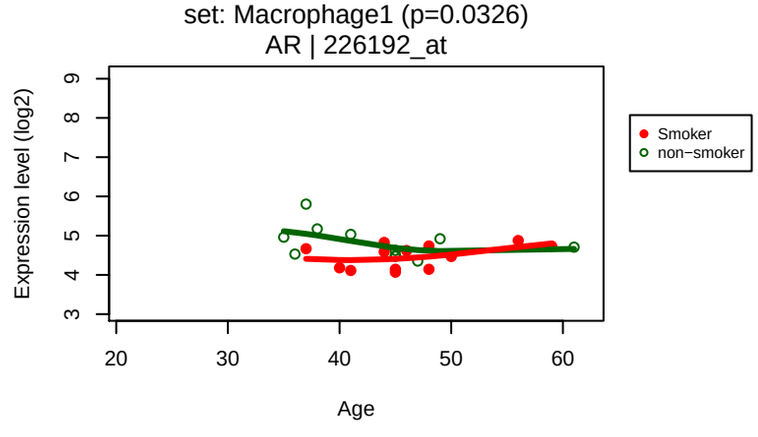

set: SmallAirway1 (p=0.7575)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker

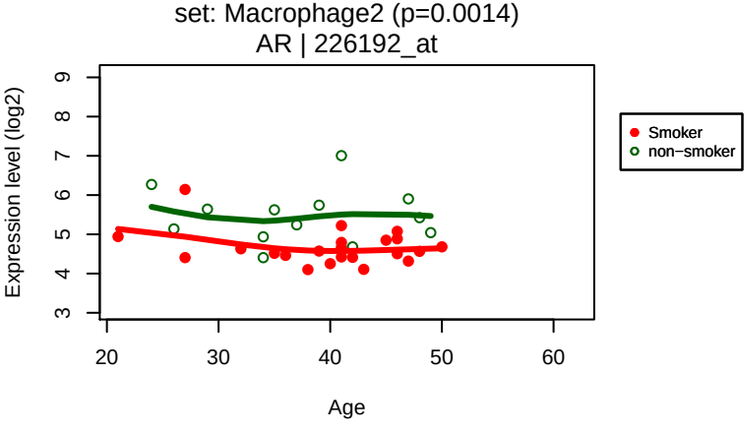

set: SmallAirway2 (p=2e−04)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker

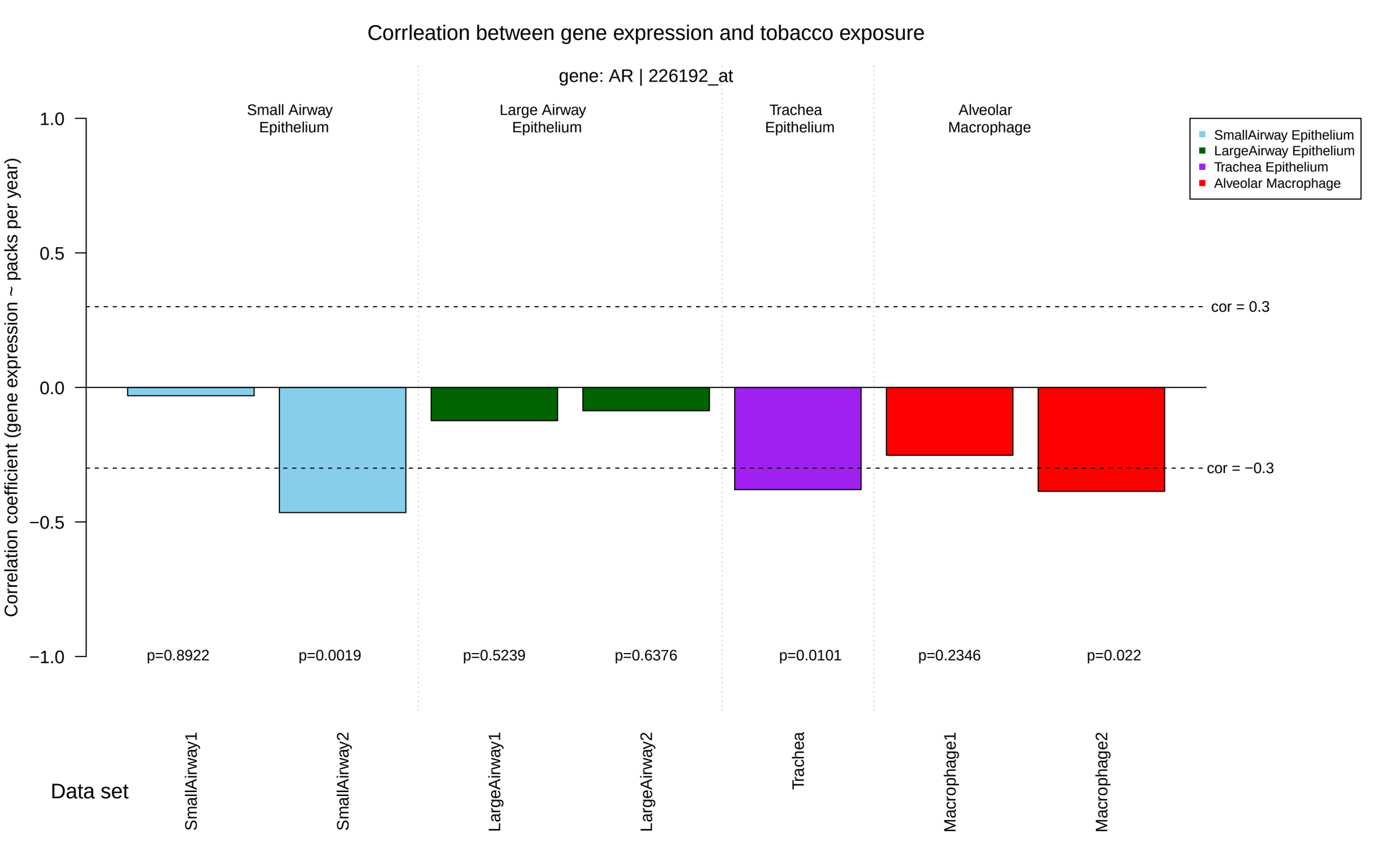

set: LargeAirway1 (p=0.5604)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0905)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0054)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0326)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0014)
AR | 226192_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AR | 226192_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8922
p=0.0019
p=0.5239
p=0.6376
p=0.0101
p=0.2346
p=0.022
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2